EMERGING ETHICS: TECHNOLOGY, AI AND ENVIRONMENT

AI ADMINISTRATIVE DECISION CHAIN → BIAS CONTROL ACROSS THE → CONTESTABILITY AND REMEDY → PRIVACY GOVERNANCE CHAIN → PUTTASWAMY PROPORTIONALITY TEST → SOCIAL-MEDIA ETHICS MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ● SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artifacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g1 and all prior artifacts unchanged

00

STAGE 00 — AI ADMINISTRATIVE DECISION CHAIN

AI ADMINISTRATIVE DECISION CHAIN | CENTRAL FOCUS | 1. AI ADMINISTRATIVE DECISION CHAIN | [1] DEFINE LAWFUL PUBLIC PURPOSE | [2] CHECK DATA FITNESS | [3] GENERATE ADVISORY OUTPUT | [4] SHOW DECISIVE FACTORS | [5] APPLY HUMAN JUDGMENT

CORE CLAIM
- CENTRAL FOCUS
- 1. AI administrative decision chain
- [1] Define lawful public purpose
- [2] Check data fitness

MECHANISM
- [3] Generate advisory output
- [4] Show decisive factors
- [5] Apply human judgment
- [6] Record independent reasons

CONSEQUENCE / LIMIT
- [7] Offer appeal and correction
- [8] Audit outcomes and drift
- VERDICT → AI may inform power; an answerable institution must exercise it.
- ANSWER USE → Use for 2021 Q2(a) and 2024 Q1(a).

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → AI may inform power; an answerable institution must exercise it.

01

STAGE 01 — BIAS CONTROL ACROSS THE LIFECYCLE

BIAS CONTROL ACROSS THE LIFECYCLE | CENTRAL FOCUS | 2. BIAS CONTROL ACROSS THE LIFECYCLE | [1] TEST REPRESENTATION IN DATA | [2] REVIEW LABELS AND PROXIES | [3] QUESTION DESIGN OBJECTIVE | [4] RED-TEAM UNEQUAL IMPACT | [5] CHECK DEPLOYMENT CONDITIONS | [6] MONITOR SUBGROUP OUTCOMES

CENTRAL FOCUS → 2. Bias control across the lifecycle → [1] Test representation in data → [2] Review labels and proxies → [3] Question design objective

CORE CLAIM
- CENTRAL FOCUS
- 2. Bias control across the lifecycle
- [1] Test representation in data
- [2] Review labels and proxies

MECHANISM
- [3] Question design objective
- [4] Red-team unequal impact
- [5] Check deployment conditions
- [6] Monitor subgroup outcomes

CONSEQUENCE / LIMIT
- [7] Break feedback loops
- [8] Correct and retrain safely
- VERDICT → Bias is a socio-technical lifecycle risk, not merely a coding defect.
- ANSWER USE → Use to replace generic claims that AI is biased.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Bias is a socio-technical lifecycle risk, not merely a coding defect.

02

STAGE 02 — CONTESTABILITY AND REMEDY

CONTESTABILITY AND REMEDY | CENTRAL FOCUS | 3. CONTESTABILITY AND REMEDY | [1] NOTIFY MATERIAL AUTOMATION | [2] GIVE INTELLIGIBLE REASONS | [3] LET CITIZEN INSPECT DATA | [4] ENABLE FACTUAL CORRECTION | [5] PROVIDE SIMPLE CHALLENGE | [6] REQUIRE HUMAN RECONSIDERATION

CENTRAL FOCUS

3. Contestability and remedy → [1] Notify material automation → [2] Give intelligible reasons

CORE CLAIM
- CENTRAL FOCUS
- 3. Contestability and remedy
- [1] Notify material automation
- [2] Give intelligible reasons

MECHANISM
- [3] Let citizen inspect data
- [4] Enable factual correction
- [5] Provide simple challenge
- [6] Require human reconsideration

CONSEQUENCE / LIMIT
- [7] Order correction or remedy
- [8] Feed lessons into redesign
- VERDICT → A right without a usable challenge route leaves automated power unchecked.
- ANSWER USE → Use for explainability, due process and accountability.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → A right without a usable challenge route leaves automated power unchecked.

03

STAGE 03 — PRIVACY GOVERNANCE CHAIN

PRIVACY GOVERNANCE CHAIN | CENTRAL FOCUS | 4. PRIVACY GOVERNANCE CHAIN | [1] STATE SPECIFIC PURPOSE | [2] ESTABLISH LAWFUL BASIS | [3] SEEK MEANINGFUL CONSENT | [4] COLLECT MINIMUM DATA | [5] SECURE AND LIMIT ACCESS | [6] RESTRICT REUSE AND RETENTION

CORE CLAIM
- CENTRAL FOCUS
- 4. Privacy governance chain
- [1] State specific purpose
- [2] Establish lawful basis

MECHANISM
- [3] Seek meaningful consent
- [4] Collect minimum data
- [5] Secure and limit access
- [6] Restrict reuse and retention

CONSEQUENCE / LIMIT
- [7] Enable correction and exit
- [8] Remedy breach and misuse
- VERDICT → Privacy protects dignity and autonomy throughout the data lifecycle.
- ANSWER USE → Use for consent, purpose and minimisation answers.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Privacy protects dignity and autonomy throughout the data lifecycle.

04

STAGE 04 — PUTTASWAMY PROPORTIONALITY TEST

PUTTASWAMY PROPORTIONALITY TEST | CENTRAL FOCUS | 5. PUTTASWAMY PROPORTIONALITY TEST | [1] PRIVACY IS A FUNDAMENTAL RIGHT | [2] IDENTIFY LEGAL AUTHORITY | [3] NAME LEGITIMATE STATE AIM | [4] PROVE RATIONAL CONNECTION | [5] TEST LEAST-RESTRICTIVE MEANS | [6] BALANCE RIGHTS AND BENEFIT

CORE CLAIM
- CENTRAL FOCUS
- 5. Puttaswamy proportionality test
- [1] Privacy is a fundamental right
- [2] Identify legal authority

MECHANISM
- [3] Name legitimate State aim
- [4] Prove rational connection
- [5] Test least-restrictive means
- [6] Balance rights and benefit

CONSEQUENCE / LIMIT
- [7] Add procedural safeguards
- [8] Keep review and remedy open
- VERDICT → A useful aim does not by itself justify a privacy restriction.
- ANSWER USE → Use as the constitutional spine for State data use.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → A useful aim does not by itself justify a privacy restriction.

05

STAGE 05 — SOCIAL-MEDIA ETHICS MAP

SOCIAL-MEDIA ETHICS MAP | CENTRAL FOCUS | 6. SOCIAL-MEDIA ETHICS MAP | [1] EXPRESSION AND ASSOCIATION | [2] MISINFORMATION AND DEEPFAKES | [3] MANIPULATION AND ADDICTION | [4] PRIVACY AND SURVEILLANCE | [5] CYBERBULLYING AND DIGNITY | [6] POLARISATION AND AMPLIFICATION

CENTRAL FOCUS

6. Social-media ethics map → [1] Expression and association → [2] Misinformation and deepfakes

CORE CLAIM
- CENTRAL FOCUS
- 6. Social-media ethics map
- [1] Expression and association
- [2] Misinformation and deepfakes

MECHANISM
- [3] Manipulation and addiction
- [4] Privacy and surveillance
- [5] Cyberbullying and dignity
- [6] Polarisation and amplification

CONSEQUENCE / LIMIT
- [7] Moderation reasons and appeal
- [8] Verification and proportionate law
- VERDICT → Protect open debate while making hidden manipulation and preventable harm answerable.
- ANSWER USE → Use for 2023 Q12 and 2025 Q1(a).

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Protect open debate while making hidden manipulation and preventable harm answerable.

06

STAGE 06 — AI ENVIRONMENTAL FOOTPRINT

AI ENVIRONMENTAL FOOTPRINT | CENTRAL FOCUS | 7. AI ENVIRONMENTAL FOOTPRINT | [1] TRAINING ELECTRICITY DEMAND | [2] REPEATED INFERENCE DEMAND | [3] COOLING AND POWER WATER | [4] HARDWARE EMBODIED IMPACTS | [5] MINERAL AND SUPPLY-CHAIN HARM | [6] REPLACEMENT AND E-WASTE

CENTRAL FOCUS → 7. AI environmental footprint → [1] Training electricity demand → [2] Repeated inference demand → [3] Cooling and power water

CORE CLAIM
- CENTRAL FOCUS
- 7. AI environmental footprint
- [1] Training electricity demand
- [2] Repeated inference demand

MECHANISM
- [3] Cooling and power water
- [4] Hardware embodied impacts
- [5] Mineral and supply-chain harm
- [6] Replacement and e-waste

CONSEQUENCE / LIMIT
- [7] Absolute-impact disclosure
- [8] Budgets, efficiency and repair
- VERDICT → A renewable-power claim alone does not establish low-impact AI.
- ANSWER USE → Use for 2024 Q7 and lifecycle analysis.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → A renewable-power claim alone does not establish low-impact AI.

07

STAGE 07 — PRECAUTION BEFORE SCALE-UP

PRECAUTION BEFORE SCALE-UP | CENTRAL FOCUS | 8. PRECAUTION BEFORE SCALE-UP | [1] IDENTIFY SERIOUS POSSIBLE HARM | [2] MAP UNCERTAINTY AND EXPOSURE | [3] TEST REVERSIBLE PILOT | [4] COMPARE SAFER ALTERNATIVES | [5] APPLY PROPORTIONATE CONTROLS | [6] SET STOP AND FALLBACK RULES

CORE CLAIM
- CENTRAL FOCUS
- 8. Precaution before scale-up
- [1] Identify serious possible harm
- [2] Map uncertainty and exposure

MECHANISM
- [3] Test reversible pilot
- [4] Compare safer alternatives
- [5] Apply proportionate controls
- [6] Set stop and fallback rules

CONSEQUENCE / LIMIT
- [7] Monitor emerging evidence
- [8] Revise before wider deployment
- VERDICT → Rio and Vellore guide AI only by analogy, not as binding AI law.
- ANSWER USE → Use for anticipatory AI and environmental governance.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Rio and Vellore guide AI only by analogy, not as binding AI law.

08

STAGE 08 — ENVIRONMENTAL JUSTICE TEST

ENVIRONMENTAL JUSTICE TEST | CENTRAL FOCUS | 9. ENVIRONMENTAL JUSTICE TEST | [1] IDENTIFY ECOLOGICAL BENEFITS | [2] IDENTIFY POLLUTION AND LOSS | [3] MAP BURDENED COMMUNITIES | [4] INCLUDE FUTURE GENERATIONS | [5] SECURE MEANINGFUL PARTICIPATION | [6] COMPARE DISTRIBUTIONAL OPTIONS

CORE CLAIM
- CENTRAL FOCUS
- 9. Environmental justice test
- [1] Identify ecological benefits
- [2] Identify pollution and loss

MECHANISM
- [3] Map burdened communities
- [4] Include future generations
- [5] Secure meaningful participation
- [6] Compare distributional options

CONSEQUENCE / LIMIT
- [7] Assign compensation and cleanup
- [8] Provide accessible remedy
- VERDICT → A green or welfare label cannot hide unequal burdens and silenced voices.
- ANSWER USE → Use for pollution, housing and siting cases.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → A green or welfare label cannot hide unequal burdens and silenced voices.

09

STAGE 09 — SUSTAINABLE-DEVELOPMENT BRIDGE

SUSTAINABLE-DEVELOPMENT BRIDGE | CENTRAL FOCUS | 10. SUSTAINABLE-DEVELOPMENT BRIDGE | [1] DEFINE PRESENT HUMAN NEED | [2] VALUE ECOSYSTEM SERVICES | [3] PROTECT FUTURE CAPABILITY | [4] COMPARE LOWER-IMPACT ROUTES | [5] INTERNALISE POLLUTION COSTS | [6] SUPPORT A JUST TRANSITION

CENTRAL FOCUS → 10. Sustainable-development bridge → [1] Define present human need → [2] Value ecosystem services → [3] Protect future capability

CORE CLAIM
- CENTRAL FOCUS
- 10. Sustainable-development bridge
- [1] Define present human need
- [2] Value ecosystem services

MECHANISM
- [3] Protect future capability
- [4] Compare lower-impact routes
- [5] Internalise pollution costs
- [6] Support a just transition

CONSEQUENCE / LIMIT
- [7] Monitor real outcomes
- [8] Revise if limits are breached
- VERDICT → Development and ecology are jointly constrained, not automatic overrides.
- ANSWER USE → Use for climate ethics and 2025 Q8.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Development and ecology are jointly constrained, not automatic overrides.

10

STAGE 10 — BORDER ENVIRONMENTAL CLEARANCE

BORDER ENVIRONMENTAL CLEARANCE | CENTRAL FOCUS | 11. BORDER ENVIRONMENTAL CLEARANCE | [1] SPECIFY SECURITY OBJECTIVE | [2] MAP SENSITIVE ECOLOGY | [3] COMPARE ROUTE ALTERNATIVES | [4] USE LEAST-DAMAGING FEASIBLE DESIGN | [5] CONSULT ON NON-CLASSIFIED IMPACTS | [6] REVIEW SECRETS THROUGH CLEARED EXPERTS

CENTRAL FOCUS

11. Border environmental clearance → [1] Specify security objective → [2] Map sensitive ecology

CORE CLAIM
- CENTRAL FOCUS
- 11. Border environmental clearance
- [1] Specify security objective
- [2] Map sensitive ecology

MECHANISM
- [3] Compare route alternatives
- [4] Use least-damaging feasible design
- [5] Consult on non-classified impacts
- [6] Review secrets through cleared experts

CONSEQUENCE / LIMIT
- [7] Condition restoration and monitoring
- [8] Publish reasons within security limits
- VERDICT → Security warrants tailored protection, not automatic ecological exemption.
- ANSWER USE → Use for 2025 Q2(b).

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Security warrants tailored protection, not automatic ecological exemption.

11

STAGE 11 — CLIMATE FAIRNESS ARCHITECTURE

CLIMATE FAIRNESS ARCHITECTURE | CENTRAL FOCUS | 12. CLIMATE FAIRNESS ARCHITECTURE | [1] COMMON DUTY TO PROTECT CLIMATE | [2] HISTORICAL CONTRIBUTION MATTERS | [3] RESPECTIVE CAPABILITY MATTERS | [4] CURRENT EMISSIONS STILL MATTER | [5] CBDR-RC DIFFERENTIATES EFFORT | [6] FINANCE AND TECHNOLOGY ENABLE ACTION

CENTRAL FOCUS → 12. Climate fairness architecture → [1] Common duty to protect climate → [2] Historical contribution matters → [3] Respective capability matters

CORE CLAIM
- CENTRAL FOCUS
- 12. Climate fairness architecture
- [1] Common duty to protect climate
- [2] Historical contribution matters

MECHANISM
- [3] Respective capability matters
- [4] Current emissions still matter
- [5] CBDR-RC differentiates effort
- [6] Finance and technology enable action

CONSEQUENCE / LIMIT
- [7] Protect vulnerable transitions
- [8] Retain reviewable national duties
- VERDICT → Differentiated responsibility means fair action, not zero obligation.
- ANSWER USE → Use for 2024 Q2(b) and climate justice.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → Differentiated responsibility means fair action, not zero obligation.

E

EXTRA — SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | 1. ARCHITECTURE | 2. CONCEPTS AND DISTINCTIONS | 3. DETAILED TREATMENT | 4. INSTITUTIONAL AND REFORM ARCHITECTURE | 5. INDIAN APPLICATIONS AND BOUNDARY CASES | 6. LIMITATIONS AND TRADE-OFFS | 7. MUST-KNOW FACTS FOR ADVANCED PRELIMS | 8. ADVANCED PRELIMS TRAPS | 9. CURRENT-ANCHOR APPLICATION | 10. PYQ-BASED ANALYTICAL APPLICATION | 11. MAINS-READY FRAMEWORK | 12. PROBABLE QUESTIONS

OPTIONAL ETHICAL DEPTH
- 1. Architecture
- 2. Concepts and distinctions
- 3. Detailed treatment
- 4. Institutional and reform architecture

ADVANCED LIMIT / TRADE-OFF
- 5. Indian applications and boundary cases
- 6. Limitations and trade-offs
- 7. Must-Know Facts for Advanced Prelims
- 8. Advanced Prelims traps

ADMINISTRATIVE APPLICATION
- 9. Current-anchor application
- 10. PYQ-based analytical application
- 11. Mains-ready framework
- 12. Probable questions

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels